

1. Given

$$f(x) = \frac{1}{\sqrt{(2\pi)^k |\Sigma|}} e^{-\frac{1}{2}(x-\mu)^T \Sigma^{-1}(x-\mu)},$$

where $x, \mu \in \mathbb{R}^k$, Σ is a k -by- k positive definite matrix and $|\Sigma|$ is its determinant.

Show that $\int_{\mathbb{R}^k} f(x) dx = 1$.

Let $y = x - \mu \in \mathbb{R}^k$. Then

$$\int_{\mathbb{R}^k} f(x) dx = \int_{\mathbb{R}^k} \frac{1}{\sqrt{(2\pi)^k |\Sigma|}} e^{-\frac{1}{2} y^T \Sigma^{-1} y} dy$$

Since Σ is a k -by- k positive definite matrix, we have $\Sigma = LL^T$ (Cholesky decomposition), where L is invertible.

Let $z = L^{-1}y$. Then

$$y^T \Sigma^{-1} y = z^T L^T (L^T)^{-1} L^{-1} L z = z^T z,$$

and hence

$$\begin{aligned} \int_{\mathbb{R}^k} f(x) dx &= \int_{\mathbb{R}^k} \frac{1}{\sqrt{(2\pi)^k |\Sigma|}} e^{-\frac{1}{2} z^T z} |\det L| dz \\ &= \int_{\mathbb{R}^k} \frac{1}{\sqrt{(2\pi)^k}} e^{-\frac{1}{2} z^T z} dz \end{aligned}$$

$$\begin{aligned}
&= \prod_{i=1}^k \int_{-\infty}^{\infty} \frac{1}{\sqrt{(2\pi)^k}} e^{-\frac{1}{2}z_i^2} dz_i \\
&= \frac{1}{\sqrt{(2\pi)^k}} \times \prod_{i=1}^k \sqrt{2\pi} \\
&= 1. \quad \square
\end{aligned}$$

2. Let A, B be n -by- n matrices and x be a n -by-1 vector.

(a) Show that $\frac{\partial}{\partial A} \text{trace}(AB) = B^T$.

(b) Show that $x^T A x = \text{trace}(x x^T A)$.

(b) Derive the maximum likelihood estimators for a multivariate Gaussian.

(a) First, $\text{trace}(AB) = \sum_{i=1}^n (AB)_{ii} = \sum_{i=1}^n \sum_{k=1}^n A_{ik} B_{ki}$.

Consider the derivative w.r.t. A_{xy} .

$$\begin{aligned}
\frac{\partial}{\partial A_{xy}} \text{trace}(AB) &= \frac{\partial}{\partial A_{xy}} \sum_{i=1}^n \sum_{k=1}^n A_{ik} B_{ki} \\
&= B_{yx}
\end{aligned}$$

Thus, $\left[\frac{\partial}{\partial A} \text{trace}(AB) \right]_{ij} = B_{ji} = (B^T)_{ij}$.

That is, $\frac{\partial}{\partial A} \text{trace}(AB) = B^T$. $\square$

(b) First, $x^T A x = \sum_{i=1}^n \sum_{j=1}^n x_i A_{ij} x_j$.

Second, $\text{trace}(x x^T A) = \sum_{i=1}^n (x x^T A)_{ii} = \sum_{i=1}^n \sum_{j=1}^n x_i x_j A_{ji}$

Therefore, $x^T A x = \text{trace}(x x^T A)$. $\square$

(c) Let $x_1, \dots, x_N \sim N(\mu, \Sigma)$.

The likelihood function is

$$L(\mu, \Sigma) = \prod_{i=1}^N p(x_i | \mu, \Sigma).$$

The log-likelihood function is

$$\ell(\mu, \Sigma) = \ln L(\mu, \Sigma)$$

$$= -\frac{kN}{2} \ln 2\pi - \frac{N}{2} \ln |\Sigma| - \frac{1}{2} \sum_{i=1}^N (x_i - \mu)^T \Sigma^{-1} (x_i - \mu)$$

$$\stackrel{(b)}{=} -\frac{kN}{2} \ln 2\pi - \frac{N}{2} \ln |\Sigma| - \frac{1}{2} \sum_{i=1}^N \text{trace}[(x_i - \mu)(x_i - \mu)^T \Sigma^{-1}]$$

Let $(x_i - \mu)(x_i - \mu)^T = X$.

$$\frac{\partial \ell}{\partial \mu} = -\frac{1}{2} \sum_{i=1}^N \frac{\partial}{\partial X} \text{trace}(X \Sigma^{-1}) \cdot \frac{\partial X}{\partial \mu}$$

$$\stackrel{(a)}{=} -\frac{1}{2} \sum_{i=1}^N (\Sigma^{-1})^T \cdot (-2)(x_i - \mu)$$

$$= \sum_{i=1}^N (\Sigma^{-1}) \cdot (x_i - \mu) , \text{ since } (\Sigma^{-1})^T = \Sigma^{-1}.$$

$$\text{Let } \frac{\partial \ell}{\partial \mu} = 0 , \text{ we get } \hat{\mu} = \frac{1}{N} \sum_{i=1}^N x_i .$$

Next, we need to compute $\frac{\partial \ell}{\partial \Sigma}$.

$$\text{By the Jacobi's formula, } \frac{\partial}{\partial \Sigma} \ln |\Sigma| = \Sigma^{-1}.$$

$$\text{Hence, } \frac{\partial \ell}{\partial \Sigma} = -\frac{N}{2} \Sigma^{-1} + \frac{1}{2} \sum_{i=1}^N \Sigma^{-1} (x_i - \hat{\mu})(x_i - \hat{\mu})^T \Sigma^{-1}$$

$$\text{Let } \frac{\partial \ell}{\partial \Sigma} = 0 , \text{ we get } \hat{\Sigma} = \frac{1}{N} \sum_{i=1}^N (x_i - \hat{\mu})(x_i - \hat{\mu})^T .$$

$$\text{Therefore, } \begin{cases} \hat{\mu} = \frac{1}{N} \sum_{i=1}^N x_i \\ \hat{\Sigma} = \frac{1}{N} \sum_{i=1}^N (x_i - \hat{\mu})(x_i - \hat{\mu})^T \end{cases}$$